

MIDTERM EXAM—SUGGESTED SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY & AUCTIONS

FEBRUARY 23, 2025

Problem 1: The NUpiterian Puzzle

1.0.1. The set of agents has size $K = 3$.

$$\Omega = \{(S, S, S), (S, S, \neg S), (S, \neg S, S), (S, \neg S, \neg S), (\neg S, S, S), (\neg S, S, \neg S), (\neg S, \neg S, S), (\neg S, \neg S, \neg S)\}.$$

The players' information partitions are:

$$\mathcal{P}_1 = \{\{(S, S, S), (\neg S, S, S)\}, \{(S, S, \neg S), (\neg S, S, \neg S)\}, \{(S, \neg S, S), (\neg S, \neg S, S)\}, \{(S, \neg S, \neg S), (\neg S, \neg S, \neg S)\}\}$$

$$\mathcal{P}_2 = \{\{(S, S, S), (S, \neg S, S)\}, \{(S, S, \neg S), (S, \neg S, \neg S)\}, \{(\neg S, S, S), (\neg S, \neg S, S)\}, \{(\neg S, S, \neg S), (\neg S, \neg S, \neg S)\}\}$$

$$\mathcal{P}_3 = \{\{(S, S, S), (S, S, \neg S)\}, \{(S, \neg S, S), (S, \neg S, \neg S)\}, \{(\neg S, S, S), (\neg S, S, \neg S)\}, \{(\neg S, \neg S, S), (\neg S, \neg S, \neg S)\}\}$$

The above completes the description of the knowledge space in terms of information partitions. You can find an equivalent description in terms of type spaces and signalling functions.

Let $P_i : \Omega \rightarrow 2^\Omega$, $i = 1, 2, 3$ denote the information partition functions of each of the players. Recall that $\mathcal{P}_i = \{P_i(\omega)\}_{\omega \in \Omega}$.

The true state of the world is (S, S, S) since $N = K = 3$.

1.0.2. First, I record a few remarks. Note that before the announcement each player i knows $P_i((S, S, S))$, i.e., player 1 knows that the state is in $\{(S, S, S), (\neg S, S, S)\}$; player 2 knows that the state is in $\{(S, S, S), (S, \neg S, S)\}$; and player 3 knows that the state is in $\{(S, S, S), (S, S, \neg S)\}$. The only event that is common knowledge is Ω .

After the announcement, the event $\Omega \setminus \{(\neg S, \neg S, \neg S)\}$ is common knowledge, i.e., that the state of the world is not $(\neg S, \neg S, \neg S)$. However, no one knows for sure if they themselves are smart or not. Since no NUpiterian leaves after the first day, it becomes common knowledge that no NUpiterian was sure that they are smart, thus it becomes common knowledge that the state is not $\{(S, \neg S, \neg S), (\neg S, S, \neg S), (\neg S, \neg S, S)\}$. Equivalently, the event $\{(S, S, S), (S, S, \neg S), (S, \neg S, S), (\neg S, S, S)\}$ is common knowledge.

Finally, since no NUpiterian leaves after the second day, it becomes common knowledge that each NUpiterian sees two other smart NUpiterians, which implies that each NUpiterian knows that the true state of the world is (S, S, S) and they all ascend to Real Life on the third day.

1.0.3. As seen from the proof above, before the announcement, the only event that was common knowledge at the true state (S, S, S) was Ω . After the announcement, the event $\Omega \setminus \{(\neg S, \neg S, \neg S)\}$ is common knowledge at (S, S, S) . This starts a chain of reasoning from the NUpiterians that culminate in all three of them figuring out that the state is (S, S, S) and thus, that all of them are smart.

1.0.4. Let's state the result that we will attempt to prove.

Proposition. *Let K be the number of NUpiterians, and assume that N of them are smart and hardworking, where $N \leq K$. After the professor's announcement, it will take exactly N days for all the smart and hardworking NUpiterians to ascend to real life.*

Proof. We begin by describing the set of states of the world after the announcement from the professor. The set of states is the set of all configurations $\omega = (\omega_1, \dots, \omega_K)$, where each $\omega_i \in \{S, \neg S\}$, and at least one ω_i is S . These $2^K - 1$ states constitute the set of states of the world:

$$\Omega = \{\omega \in \{S, \neg S\}^K : |\{i : \omega_i = S\}| \geq 1\}.$$

Each NUpiterian's initial information partition function is given as follows: if ω is a state in which a NUpiterian different from i is smart, then

$$P_i^1(\omega) = \{(S, \omega_{-i}), (\neg S, \omega_{-i})\},$$

and if ω' is the state in which all others are smart, then

$$P_i^1(\omega') = \{\omega'\}.$$

The NUpiterians' initial information partitions are given by $\mathcal{P}_i^1 = \{P_i^1(\omega)\}_{\omega \in \Omega}$ for $i = 1, \dots, K$.

A NUpiterian i knows that she is smart or not smart if either she knows that the event $\{\omega : \omega_i = S\}$ occurs or she knows that the event $\{\omega : \omega_i = \neg S\}$ occurs. Thus, the event "i knows if she is smart or not smart" is

$$E_i = \{\omega : P_i(\omega) \subseteq \{\omega : \omega_i = S\} \text{ or } P_i(\omega) \subseteq \{\omega : \omega_i = \neg S\}\}.$$

When the state ω is such that there is exactly one NUpiterian i for whom $\omega_i = S$, then $P_i^1(\omega) \subseteq E_i$, so i realizes she is smart and ascends to Real Life.

Let $F^1 = \{\omega : |\{i : \omega_i = S\}| = 1\}$, which is the set of states in which someone ascends to Real Life on the first day. If no NUpiterians ascend to Real Life on the first day, then all NUpiterians obtain the additional information that the state is not in F^1 , which implies that for all i and all $\omega \notin F^1$, we have $P_i^2(\omega) = P_i^1(\omega) \setminus F^1$. This means that in any such state, every NUpiterian concludes that there are at least two smart NUpiterians. Then we have that $P_i^2(\omega) = P_i^1(\omega)$ unless $\omega_j = S$ for exactly one individual $j \neq i$, in which case

$$P_i^2(S, \omega_{-i}) = \{(S, \omega_{-i})\}$$

and also

$$P_j^2(S, \omega_{-j}) = \{(S, \omega_{-j})\}.$$

Thus, in any state ω for which $\omega_i = S$ and $\omega_j = S$ for precisely two NUpiterians i and j , we have that

$$P_i^2(\omega) \subseteq E_i \quad \text{and} \quad P_j^2(\omega) \subseteq E_j,$$

and hence i and j each ascend to Real Life on the second day.

Let $F^2 = \{\omega : |\{i : \omega_i = S\}| = 2\}$, which is the set of states in which NUpiterians ascend to Real Life on the second day. In states in which no NUpiterian has ascended to Real Life on the second day, i.e., $\omega \notin F^1 \cup F^2$, all NUpiterians conclude that there are at least three smart NUpiterians, and the process continues with $P_i^3(\omega) = P_i^2(\omega) \setminus F^2$. Proceeding inductively, it is easy to see that if there are N smart NUpiterians, then no one ascends to Real Life until the N -th day, when the N smart NUpiterians realize they are smart and unfortunately have to ascend to Real Life. 

Problem 2: NU Roommates

2.0.1. $N = \{1, 2\}$, $T = \{v_H, v_L\}$, $\Omega = T \times T$, and $\tau_i(v_i, v_j) = v_i$ for $i = 1, 2$. The common prior is given by $\mu(v_H, v_H) = p^2$, $\mu(v_H, v_L) = \mu(v_L, v_H) = p(1 - p)$, and $\mu(v_L, v_L) = (1 - p)^2$, which induces the posteriors as given in the problem statement.

2.0.2. Let σ be a BNE profile. Towards a contradiction, suppose that $\sigma_i(C | v_L) = 1$ for player i . Then $E[u_i(\sigma, v_L) | v_L] = v_L - c = -c < 0$. Therefore, it is optimal for player i to deviate to $\sigma_i(M | v_L) = 1$. Therefore, $\sigma_i(C | v_L) = 1$ can never be part of an equilibrium for any player. In fact, it can be seen that playing C with probability 1 when the type is low is *interim-dominated* for player i .

2.0.3. Let $\alpha_i = \sigma_i(C | v_H)$, for $i = 1, 2$. From the previous item, $\sigma_i(C | v_L) = 0$ for $i = 1, 2$. Let $\sigma_j = (\alpha_j, 0)$ denote player j 's strategy. Then:

$$E[u_i(M, (\alpha_j, 0), v_H) | v_H] = v_H - c,$$

$$E[u_i(C, (\alpha_j, 0), v_H) | v_H] = (1 - p) \cdot 0 \cdot v_H + p\alpha_j v_H = p\alpha_j v_H.$$

2.0.4. From 2.2, we know that we only need to compute the best response of player i against the strategy $(\alpha_j, 0)$, when their type is v_H since cleaning is interim-dominated by going to the movies when the type of the player is v_L . Therefore:

$$\text{BR}_i((\alpha_j, 0); v_H) = \begin{cases} \alpha_i = 1 & \text{if } v_H - c > p\alpha_j v_H, \\ \alpha_i \in [0, 1] & \text{if } v_H - c = p\alpha_j v_H, \\ \alpha_i = 0 & \text{if } v_H - c < p\alpha_j v_H. \end{cases}$$

2.0.5. Suppose that $\alpha_1 = 0$, then, from the best response above it is clear that $\alpha_2 = 1$, and if this is the case, then $\alpha_1 = 0$ since $v_H - c < p v_H$ from our initial assumption that $v_H(1 - p) < c$. Then, $\alpha_1 = 0$, i.e., player 1 never cleans, and $\alpha_2 = 1$, i.e., player 2 cleans when high is an equilibrium.

The same analysis shows that $\alpha_2 = 0$, i.e., player 2 never cleans, and $\alpha_1 = 1$, i.e., player 1 cleans when high is also an equilibrium.

There are no other pure strategy equilibria.

2.0.6. Suppose that player i of the high type is indifferent between cleaning and not cleaning, then:

$$v_H - c = p\alpha_j v_H \Rightarrow \alpha_j = \frac{v_H - c}{p v_H}.$$

Notice that α_j is well-defined, since by the parametric assumptions, $0 < \alpha_j < 1$. If player j mixes according to α_j , then he must also be indifferent, and thus $\alpha_j = \alpha_i$. This constitutes an equilibrium in which both players of type high play a completely mixed behavioral strategy.

2.0.7. Let A denote the event “there is at least one roommate with value v_H (values a clean room), but the room is not cleaned.” Consider first the pure strategy equilibrium in which $\alpha_1 = 0$, $\alpha_2 = 1$, i.e., roommate 1 never cleans, and roommate 2 cleans only when the type is high. Then, notice that whenever player 2 has a high type, the room is cleaned, which happens with probability $p + p(1 - p)$. On the other hand, if both of them are low type, the room doesn’t get cleaned because nobody values it, which happens with probability $(1 - p)^2$. Finally, the only case in which the room is not cleaned but at least one roommate values it is when roommate 1 has a high type, and roommate 2 has a low type, which happens with probability $p(1 - p)$. Thus, we conclude that for this equilibrium $\Pr(A) = p(1 - p)$. A similar analysis shows that for the equilibrium in which $\alpha_1 = 1$ and $\alpha_2 = 0$ we also get $\Pr(A) = p(1 - p)$.

The analysis for the mixed strategy equilibrium is slightly more involved. Let α^* denote the symmetric completely mixed action equilibrium we found in item 2.6 above. There are three cases in which at least one roommate has a high type: (v_H, v_H) , (v_H, v_L) , and (v_L, v_H) , each with probability p^2 , $p(1 - p)$, and $(1 - p)p$ respectively. Conditional on the type profile, the probability that the room is not cleaned is:

- If (v_H, v_H) : the room is not cleaned with probability $(1 - \alpha^*)^2$.
- If (v_H, v_L) : the room is not cleaned with probability $(1 - \alpha^*)$.
- If (v_L, v_H) : the room is not cleaned with probability $(1 - \alpha^*)$.

Thus, $\Pr(A) = p^2(1 - \alpha^*)^2 + 2p(1 - p)(1 - \alpha^*)$.

2.0.8. (a) Notice that now when both roommates clean, the cost of cleaning is shared equally among them. We can interpret this cost decrease as a result of dividing the work equally among them, and thus, only paying half the cost. When only one roommate cleans, the payoffs remain the same as in the first part of this problem. Similarly for the remaining action profile configurations.

(b) Notice that irrespective of the strategy $\sigma_j : T \rightarrow \Delta(\{C, M\})$ of player j ,

$$E[u_i(C, \sigma_j, v_L) \mid v_L] < 0 \leq E[u_i(M, \sigma_j, v_L) \mid v_L].$$

Therefore, cleaning is interim-dominated by going to the movies when the type is low. Therefore, any equilibrium will have that $\sigma_i(C \mid v_L) = 0$ for $i = 1, 2$.

Now, as before, define $\alpha_i = \sigma_i(C \mid v_H)$ for $i = 1, 2$. Let $\sigma_j = (\alpha_j, 0)$ denote player j ’s strategy. Then:

$$\begin{aligned} E[u_i(M, (\alpha_j, 0), v_H) \mid v_H] &= p \left[\alpha_j \left(v_H - \frac{c}{2} \right) + (1 - \alpha_j)(v_H - c) \right] + (1 - p)(v_H - c) \\ &= v_H - \frac{c}{2}(2 - \alpha_j p), \\ E[u_i(C, (\alpha_j, 0), v_H) \mid v_H] &= p\alpha_j v_H. \end{aligned}$$

Therefore, the best response of player i against $(\alpha_j, 0)$ when her type is v_H is:

$$\text{BR}_i((\alpha_j, 0); v_H) = \begin{cases} \alpha_i = 1 & \text{if } v_H - \frac{c}{2}(2 - \alpha_j p) > p\alpha_j v_H, \\ \alpha_i \in [0, 1] & \text{if } v_H - \frac{c}{2}(2 - \alpha_j p) = p\alpha_j v_H, \\ \alpha_i = 0 & \text{if } v_H - \frac{c}{2}(2 - \alpha_j p) < p\alpha_j v_H. \end{cases}$$

With the best response correspondence, we are ready to determine the equilibria. We start with the pure strategy equilibria.

First we check if the profiles in which one of the roommates never cleans while the other roommate cleans when high type can be equilibria. Suppose that $\alpha_1 = 0$. Then the best response of roommate 2 is $\alpha_2 = 1$. For $\alpha_1 = 0$ to be a strict best response to $\alpha_2 = 1$, we need

$$v_H - \frac{c}{2}(2 - p) < p\alpha_j v_H \Rightarrow \frac{v_H - c}{v_H - c/2} < p.$$

We obtain the same condition for the profile $\alpha_1 = 1$ and $\alpha_2 = 0$. Joining the previous condition with our parametric assumptions yields:

$$\frac{v_H - c}{v_H - c/2} < p < 1 \quad (1)$$

Thus, if condition (1) holds, $((0, 0), (1, 0))$ and $((1, 0), (0, 0))$ are pure strategy equilibrium profiles.

Now, note that a new “class” of equilibrium appears: high types always clean. To see this, suppose that $\alpha_1 = 1$. For $\alpha_2 = 1$ to be a strict best response, we need that:

$$v_H - \frac{c}{2}(2 - p) > p\alpha_j v_H \Rightarrow \frac{v_H - c}{v_H - c/2} > p.$$

We obtain the same condition for $\alpha_1 = 1$ to be a best response against $\alpha_2 = 1$. Joining the previous condition with our parametric assumptions yields:

$$\frac{v_H - c}{v_H} < p < \frac{v_H - c}{v_H - c/2}. \quad (2)$$

Thus, if condition (2) holds, $((1, 0), (1, 0))$ is an equilibrium profile.

Finally, we analyze the possibility of a completely mixed equilibrium. Suppose that player i of the high type is indifferent between cleaning and not cleaning, then:

$$v_H - \frac{c}{2}(2 - \alpha_j p) = p\alpha_j v_H \Rightarrow \alpha_j = \frac{v_H - c}{p(v_H - c/2)}.$$

To check that α_j is well-defined as a completely mixed equilibrium, we need to check that $0 < \alpha_j < 1$. Our parametric assumptions guarantee that $\alpha_j > 0$, but in order to have $\alpha_j < 1$ the following condition must hold:

$$\frac{v_H - c}{v_H - c/2} < p.$$

The above together with the remaining parametric assumptions are equivalent to condition (1). The same analysis concludes that if j mixes with according to α_j , then he must also be indifferent and thus $\alpha_j = \alpha_i$. Thus, if condition (1) holds, we have an equilibrium in which both players of the high type play a completely mixed behavioral strategy.

There is one subtle case that we still need to consider, which is the boundary case in which the parameters are such that $\frac{v_H - c}{p(v_H - c/2)} = 1$. This of course implies that

$$p = \frac{v_H - c}{v_H - c/2}. \quad (3)$$

If we substitute the value of p given by equation (3) into the best response correspondence of player i we obtain after some tedious algebra:

$$\text{BR}_i((\alpha_j, 0); v_H) = \begin{cases} \alpha_i = 1 & \text{if } \alpha_j < 1, \\ \alpha_i \in [0, 1] & \text{if } \alpha_j = 1. \end{cases}$$

It is easy to see that, in this case, we obtain a continuum of equilibria. Specifically, for any pair $(\alpha_1, \alpha_2) \in \{(x, y) \in [0, 1]^2 \mid x = 1 \text{ or } y = 1\}$, these values represent the probabilities with which high types choose to clean, while all low types, as previously analyzed, never clean.

(c) We conclude that the main differences compared to the original case are: (1) asymmetric equilibria, in which only one high type cleans while the other does not, only appear when the probability of high types is sufficiently large; (2) a new symmetric equilibrium emerges, in which both high types clean, if the probability of high types is sufficiently low; (3) a completely mixed equilibrium still exists in the same region where the asymmetric equilibria appear; and (4) there is a boundary case, where, for a specific value of the probability of high types, a continuum of equilibria exists.

Problem 3: Sharing information

3.0.1. True. Follows immediately from theorem 1.30 in the slides (Geanakoplos and Polemarchakis, 1982).

3.0.2. False. Consider the example from the post-lecture quiz on 04/02/25. Let $\mu(\omega) = 1/12$ for all $\omega \in \Omega$ and $A = \{1, 2, 3\}$. Player 1 ends up with a coarser information partition than player 2, which ends up with the same information partition as she originally started with.

Problem 4: A Proverb About Knowledge

4.0.1. Let us consider the agents in order and first examine what each part of the proverb implies for each agent.

Fool The fool, at some state of the world, does not know an event, and at that state, the fool also fails to know that he does not know. This means that there exists an event B and $\omega \in B$ such that $\omega \notin L_F B$, or equivalently, $\omega \in (L_F B)^c$ and $\omega \notin L_F((L_F B)^c)$. Thus, $(L_F B)^c \neq L_F((L_F B)^c)$, which means the fool violates the negative introspection axiom.

Simple The simple agent, at some state of the world, does not know an event, but at that state, he knows that he does not know. This means that there exists an event B and $\omega \in B$ such that $\omega \in (L_S B)^c$ and $\omega \in L_S((L_S B)^c)$. If this is true for all events B , i.e., $L_S B = L_S((L_S B)^c)$, then the simple agent does not violate the negative introspection axiom. If this does not hold for all events B , then the simple agent can fail to satisfy the negative introspection axiom.

Asleep The asleep agent, at some state of the world, knows an event, but at that same state, he fails to know that he knows. This means that there exists an event B and $\omega \in B$ such that $\omega \in L_A B$ but $\omega \notin L_A(L_A B)$, or equivalently, $\omega \in (L_A(L_A B))^c$. Thus, $L_A(L_A B) \neq L_A B$, which means the asleep agent violates the positive introspection axiom.

Wise The wise agent, at some state of the world, knows an event and, at that same state, knows that he knows it. This means that there exists an event B and $\omega \in B$ such that $\omega \in L_W B$ and $\omega \in L_W(L_W B)$. If this is true for all events B , i.e., $L_W B = L_W(L_W B)$, then the wise agent does not violate the positive introspection axiom. If this does not hold for all events B , then the wise agent can fail to satisfy the positive introspection axiom.

4.0.2. From the previous item, it is clear that for both the fool and the asleep, there exists an event for which there is a state of the world at which they fail the negative introspection and positive introspection axioms, respectively. Therefore, the knowledge operators of the fool and the asleep cannot come from a standard model with information partitions.

On the other hand, the simple and the wise can come from a standard knowledge model with information partitions, per Theorem 1.7 in the slides and the remark that follows it, provided that their respective knowledge operators satisfy all of Kripke's axioms.

4.0.3. Let $A = \{\omega_3, \omega_4\}$. Note that $KA = \{\omega_3, \omega_4\}$, $(KA)^c = \{\omega_1, \omega_2\}$ and $K((KA)^c) = \{\omega_1\}$. Therefore, we can say that at ω_2 the agent is a *fool* at that state he doesn't know A but he doesn't know that he doesn't know A . At ω_1 , the agent is *simple* since he doesn't know A but he knows that he doesn't know A .

Now let $B = \{\omega_1, \omega_2\}$. Note that $KB = \{\omega_1, \omega_2\}$ and $K(KB) = \emptyset$. Then at both ω_1 and ω_2 this agent is *simple* since knows B in both states, but also in both he doesn't know that he knows B .

Finally, let $C = \{\omega_3, \omega_4\}$. Note that $KC = \{\omega_3, \omega_4\}$ and $K(KC) = \{\omega_3, \omega_4\}$. Thus, at both ω_3 and ω_4 the agent is *wise*, since he knows C in both and knows that he knows in both.

From the previous examples, notice that being a fool, being simple, being asleep, and being wise are not mutually exclusive, as you can easily check.